Nordic Flow

Goods allocation across existing warehouses

A Sweden-first proposal for Company A

Round 1 proposal / 26 October 2026

Why fulfilment allocation comes first

Sweden is the proposed entry market. Week 1 must confirm the priority.

Opportunity	Why it matters	Trade-off for the first MVP
Goods allocation	Direct link to fulfilment cost and service	Requires usable stock, rates and capacity data
Manual task automation	Can reduce employee effort	The case gives no task volumes or effort baseline
Store process standardisation	Can improve consistency	Local workflows need broader discovery
Group-wide systems replacement	Could reduce fragmentation	Wide migration scope delays the first test

Decision: test a bounded allocation service before committing to a wider programme.

Case pp. 9–12. Prioritisation is the team's judgement, subject to discovery.

The allocation decision

Which source and existing warehouse should serve each confirmed order?

What the case establishes

Company A sells building materials across Denmark, Norway and Sweden.

Fragmented systems and fulfilment issues affect efficiency and reliability.

What we need to validate

Planners may optimise individual legs without seeing total cost.

Rates, stock and handling limits may sit in separate local systems.

Industry signal Byggmax's 2025 report describes logistics restructuring to improve flow control, planning and delivery efficiency.

Case pp. 8–11. Byggmax Annual Report 2025, p. 7. Industry analogy, not evidence about Company A.

A focused pilot in one Swedish region

Proposed scope assumptions. Company A must confirm sites and data availability.

2 suppliers

3 existing warehouses

20 palletised SKUs

One week per planning run

Included

Allocate inbound supply and fulfil confirmed orders.
Compare feasible plans and explain cost differences.
Planner approval before execution.

Boundaries

Existing warehouses only.
Vehicle schedules remain in the current TMS.
Later phases can add direct delivery and transfers.

Initial exclusions: new warehouses, vehicle routing, demand forecasting, production planning and cross-border flows.

The planner stays in control

A recommendation file fits the existing operating process.

01 Load

Orders, available supply, rates and warehouse limits

02 Compare

Current allocation and a feasible lower-cost plan

03 Review

Exceptions, service rules and cost by leg

04 Approve

Planner exports the plan. Current systems execute.

Recommendation output

Supplier / SKU	Warehouse	Destination	Quantity / required date
S1 / SKU-01	Warehouse A	Order zone North	60 pallets / within the week
S2 / SKU-01	Warehouse B	Order zone South	40 pallets / within the week

Illustrative output only. Each run retains input version, approvals and planner overrides.

A constrained optimisation service

A shared decision model can work alongside local ERP, WMS and TMS systems.

Data and integration

Local CSV / API adapters

Validation and unit conversion

Scenario store and audit history

Later AI option: SKU mapping suggestions, with human validation.

Decision engine

Minimise inbound freight + handling + outbound freight + valid storage fees

Meet demand and flow balance.
Respect available supply, SKU eligibility, warehouse handling limits and allowed lanes.

LP for divisible quantities. MILP for whole pallets and discrete commercial rules.

Technical reference: Google OR-Tools, Mixed Integer Programming. Solver time limits and status remain visible.

A transparent allocation example

Synthetic data. Cost units per pallet. Both plans serve the same 100 pallets.

Source	A / North	A / South	B / North	B / South
S1	5	9	11	8
S2	6	10	8	5

Supply S1 / S2 = 60 / 40. Demand North / South = 60 / 40. Capacity A / B = 100 / 60.

700

Preferred-warehouse baseline

500

Lowest feasible cost

Plan	Allocation	Total cost
Baseline	60 via S1–A–North + 40 via S2–A–South	700
Optimised	60 via S1–A–North + 40 via S2–B–South	500

Synthetic reduction: 200 cost units (28.6%). This illustrates the mechanism and is not a forecast for Company A.

Pilot success measures

Proposed acceptance gates. Targets are hypotheses until Company A validates them.

Measure	Proposed pilot gate	How we verify it
Relevant logistics cost	At least 5% below a feasible baseline	Replay the same orders and constraints
Planner preparation time	At least 30% lower	Observe the full task, including data checks
Customer service	No deterioration in on-time, in-full rate	Compare matched live orders after planner review
Feasibility and data	Zero hard-constraint violations	Validate all required fields and each plan

Commercial decision

Scale only when validated savings cover delivery and operating costs. Assess labour savings separately to avoid double counting.

Cost saving = (baseline cost − candidate cost) / baseline cost. No revenue uplift or annual ROI assumed.

One allocation service, different local rollouts

Country differences change configuration, integration and adoption.

Market	Proposed rollout approach	Local conditions to confirm
Sweden first	Pilot one region using CSV exports. Planner reviews every recommendation.	Rates, delivery promises, SKU handling and available warehouse capacity
Denmark next	Reuse the engine. Adapt the import process to reduce manual preparation effort.	Local data exports, process ownership and the priority of allocation vs other tasks
Norway after readiness	Validate branch and distribution workflows before connecting the shared service.	Store receiving limits, carrier rules and country-specific system interfaces

Shared core

Common data contract, optimisation rules and audit trail.
Local teams own rates, service rules and approval rights.

Case pp. 10–12. Country order and implementation choices are proposals, not findings about local systems.

An eight-week path to a pilot decision

Indicative plan after sponsor approval and access to usable data.

Timing	Work and accountable role	Exit condition
Week 1	Product lead + local planner validate problem and data	Agreed baseline, scope and service rules
Weeks 2–4	Optimisation engineer + application/data engineer build MVP	Auditable plans pass historical replay
Weeks 5–6	Local operations sponsor runs a controlled trial	Planner acceptance and service evidence
Weeks 7–8	Sponsor reviews economics and rollout readiness	Scale, revise or stop against agreed gates

Main risks

Poor rates or capacity data: block invalid runs.
Low adoption: co-design with planners and retain overrides.

Requested decision: sponsor a bounded Swedish pilot and nominate a local process owner.